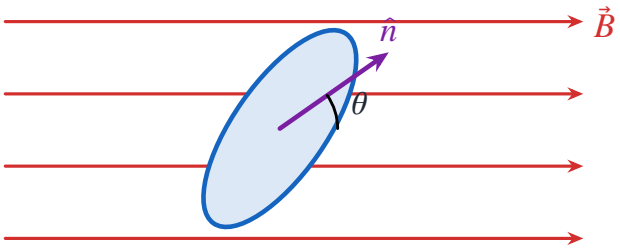
ELECTROMAGNETIC INDUCTION

Flux • Faraday • Lenz

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1 Magnetic Flux

$$\Phi = \vec{B} \cdot \vec{A} = BA\cos\theta$$



θ is between $\vec{B}$ and the area's normal. Unit weber (T m^2).

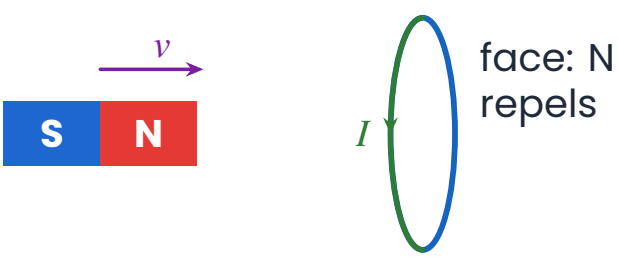
2 Faraday's Law

$$\varepsilon = -N\frac{d\Phi}{dt}$$

Change B, A or θ, any one: an emf appears. Steady flux gives none.

3 Lenz's Law

I_{ind} opposes $\Delta\Phi$



An approaching N pole makes the near face N: it repels. Energy is conserved.

4 Induced Charge

$$q = \frac{N\,\Delta\Phi}{R}$$

Independent of how fast the flux changes: only the total change counts.

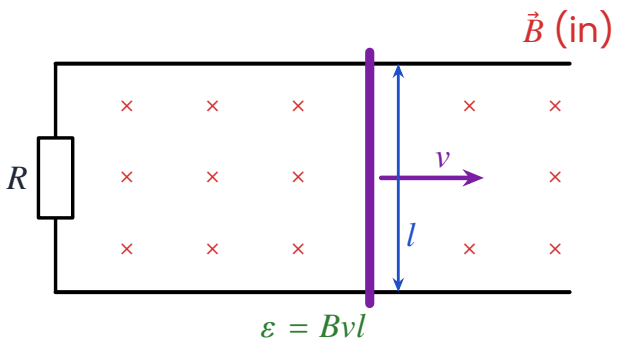
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Motional EMF

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5 Rod on Rails

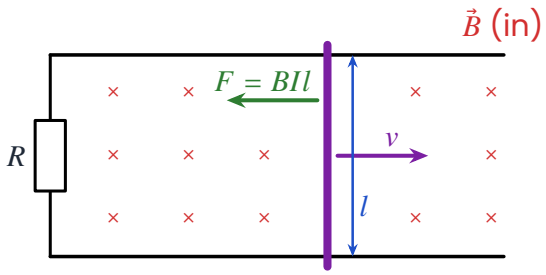
$$\varepsilon = Bvl$$



$B = 0.4$ T, $l = 0.5$ m, $v = 5$ m/s: $\varepsilon = 1$ V.

6 Current and Brake

$$I = \frac{Bvl}{R}, \qquad F = \frac{B^2 l^2 v}{R}$$



With $R = 2\ \Omega$: $I = 0.5$ A, $F = 0.1$ N, opposite to v.

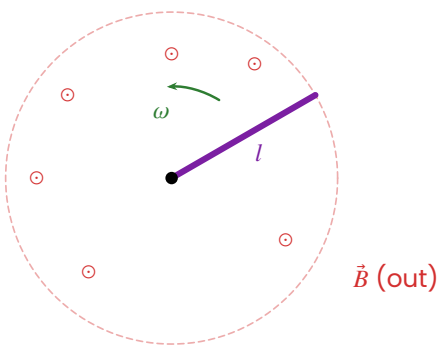
7 Terminal Velocity

$$v_t = \frac{mgR}{B^2 l^2}$$

A rod falling on vertical rails stops speeding up when the brake equals its weight.

8 Rotating Rod

$$\varepsilon = \tfrac{1}{2} B\omega l^2$$



A disc of radius R: $\frac{1}{2} B\omega R^2$.

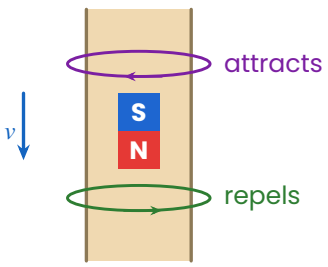
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Eddy Currents • AC Generator

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9 Eddy Currents

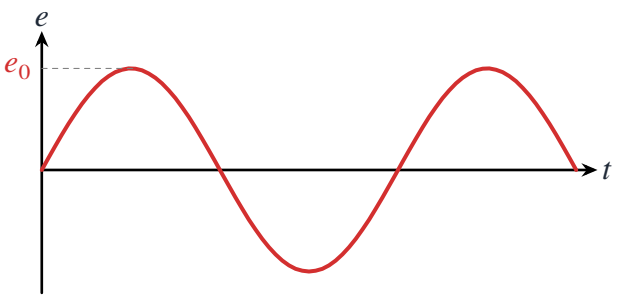
$$F_{\text{drag}} = k\,v \;\Rightarrow\; v_t = \frac{mg}{k}$$



A magnet falls slowly in a copper pipe; also used in brakes and induction cookers.

10 AC Generator

$$e = NBA\,\omega \sin \omega t$$



Peak $e_0 = NBA\omega$. $N = 100$, $B = 0.1$ T, $A = 100$ cm^2, $\omega = 314$: $e_0 = 31.4$ V.

11 Energy Check

$$P = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R$$

The work against the brake appears as heat in the circuit.

12 Lenz in One Line

$$\Phi \uparrow \Rightarrow \vec{B}_{\text{ind}} \text{ opposite}$$

Flux falling: induced B along the original. The coil always resists the change.

ELECTROMAGNETIC INDUCTION

Self and Mutual Inductance

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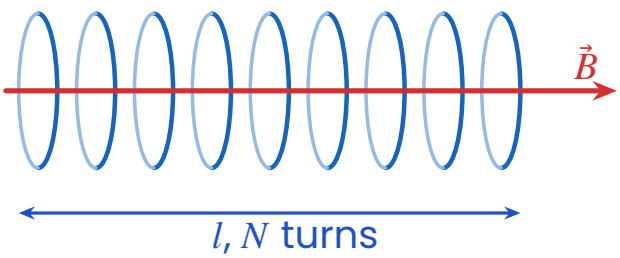
13 Self Inductance

$$N\Phi = LI, \qquad \varepsilon = -L\frac{dI}{dt}$$

A coil fights changes in its own current: electrical inertia.

14 Long Solenoid

$$L = \frac{\mu_0 N^2 A}{l}$$



$N = 1000$, $l = 0.5$ m, $A = 10$ cm^2: $L = 2.51$ mH.

15 Mutual Inductance

$$M = \frac{\mu_0 N_1 N_2 A}{l}, \qquad \varepsilon_2 = -M\frac{dI_1}{dt}$$

$M_{12} = M_{21}$; with leakage, $M = k\sqrt{L_1 L_2}$, $k \leq 1$.

16 Energy in an Inductor

$$U = \tfrac{1}{2}LI^2, \qquad u = \frac{B^2}{2\mu_0}$$

Stored in the magnetic field, as energy per volume u.

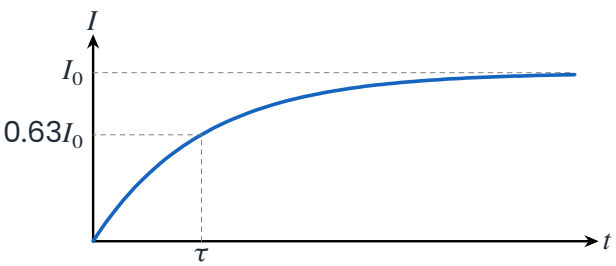
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LR Circuits

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17 Growth

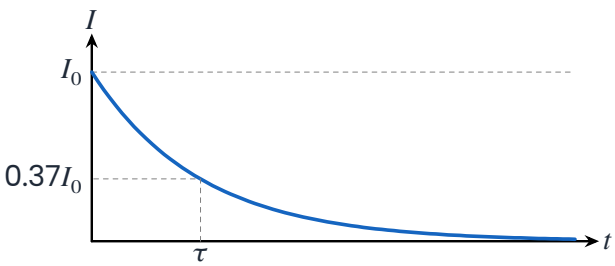
$$I = I_0\left(1 - e^{-t/\tau}\right), \quad \tau = \frac{L}{R}$$



$L = 2$ H, $R = 4\ \Omega$, $E = 12$ V: $\tau = 0.5$ s, $I_0 = 3$ A.

18 Decay

$$I = I_0\, e^{-t/\tau}$$



After one τ, 37 % is left; after 5τ, almost none.

19 At the Switch

$$t = 0:\ L \to \text{open}$$
$$t \to \infty:\ L \to \text{wire}$$

Current through an inductor cannot jump.

20 Quick Numerical

A 0.5 m rod moves at 5 m/s across 0.4 T, circuit resistance 2 Ω. Find I and the force.

$$\varepsilon = Bvl = 1\ \text{V}, \quad I = 0.5\ \text{A}$$

$$F = BIl = 0.1\ \text{N}$$

ELECTROMAGNETIC INDUCTION

Quick Revision • Traps

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21 Every Formula at a Glance

Situation	Result
Flux, Faraday	$\Phi = BA\cos\theta$, $\varepsilon = -N\,d\Phi/dt$
Rod on rails / rotating rod	Bvl / $\frac{1}{2}B\omega l^2$
Induced charge	$q = N\Delta\Phi/R$
AC generator	$e = NBA\omega\sin\omega t$
Solenoid, energy	$L = \mu_0 N^2 A/l$, $U = \frac{1}{2}LI^2$
LR circuit	$\tau = L/R$; 63 % / 37 % at $t = \tau$

22 Common Traps

- θ in $\Phi = BA\cos\theta$ is from the normal, not the plane.
- Steady flux, however large, induces nothing.
- Induced charge does not depend on the time taken.
- Lenz: the induced current opposes the change, not the field.

23 Units

Quantity	Unit
Φ	Wb = T m^2
L, M	H = Wb/A
$\tau = L/R$	s
μ_0	$4\pi \times 10^{-7}$ H/m